

Instructions

This mock mirrors the Tredence data-science assessment: aptitude, verbal, statistics, ML and CS MCQs plus SQL and Python.

Attempt in one ~2-hour sitting. This is Mock 1 of 3; each uses a different question set.

Section 1 - Aptitude & Reasoning

Q1. 12.5% as a fraction is:

(A) $\frac{1}{6}$

(B) $\frac{1}{8}$ ✓

(C) $\frac{1}{4}$

(D) $\frac{1}{5}$

Answer: (B) $\frac{1}{8}$

Authenticity: Community Reported

Q2. Average of 10, 20, 30, 40, 50:

(A) 25

(B) 30 ✓

(C) 35

(D) 40

Answer: (B) 30

Authenticity: Community Reported

Q3. A 200 m train crosses a 300 m platform at 20 m/s in:

(A) 20 s

(B) 25 s ✓

(C) 30 s

(D) 15 s

Answer: (B) 25 s

Authenticity: Research-Backed Company Style

Q4. Next term: 4, 9, 16, 25, ?

(A) 30

(B) 36 ✓

(C) 49

(D) 35

Answer: (B) 36

Authenticity: Frequently Repeated

Q5. Odd one out: 3, 5, 7, 9, 11.

(A) 5

(B) 7

(C) 9 ✓

(D) 11

Answer: (C) 9

Authenticity: Community Reported

Q6. All engineers are graduates. Some graduates are managers. Conclusion: Some engineers are managers.

(A) Follows

(B) Does not follow ✓

(C) Definitely true

(D) None

Answer: (B) Does not follow

Authenticity: Frequently Repeated

Section 2 - Verbal

Q1. Synonym of METICULOUS:

(A) Careless

(B) Careful ✓

(C) Quick

(D) Rough

Answer: (B) Careful

Authenticity: Community Reported

Q2. Antonym of RIGID:

(A) Stiff

(B) Flexible ✓

(C) Hard

(D) Firm

Answer: (B) Flexible

Authenticity: Community Reported

Q3. Passage: 'Data-driven organisations use analytics to guide decisions rather than relying on intuition alone. By measuring outcomes and testing hypotheses they reduce guesswork. However, poor data quality can lead to misleading conclusions.' | Main advantage of being data-driven?

(A) Removes need for data

(B) Guides decisions and reduces guesswork ✓

(C) Relies on intuition

(D) Increases guesswork

Answer: (B) Guides decisions and reduces guesswork

Authenticity: Research-Backed Company Style

Section 3 - Statistics & Probability

Q1. Find the mean of 2, 4, 6, 8, 10.

(A) 5

(B) 6 ✓

(C) 7

(D) 8

Answer: (B) 6

Authenticity: Community Reported

Q2. Standard deviation is a measure of:

(A) Central tendency

(B) Dispersion ✓

(C) Correlation

(D) Skewness

Answer: (B) Dispersion

Authenticity: Frequently Repeated

Q3. In a right-skewed distribution, the usual order is:

(A) Mean < Median < Mode

(B) Mean > Median > Mode ✓

(C) Mean = Median = Mode

(D) Median > Mean > Mode

Answer: (B) Mean > Median > Mode

Authenticity: Community Reported

Q4. If $P(A)=0.3$, $P(B)=0.5$, independent, then $P(A \text{ and } B)$ is:

(A) 0.8

(B) 0.15 ✓

(C) 0.2

(D) 0.5

Answer: (B) 0.15

Authenticity: Community Reported

Q5. Two fair coins tossed; $P(\text{exactly one head})$:

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$ ✓

(C) $\frac{3}{4}$

(D) $\frac{1}{3}$

Answer: (B) $\frac{1}{2}$

Authenticity: Community Reported

Q6. Bayes' theorem is used to:

(A) Compute variance

(B) Relate $P(A|B)$ to $P(B|A)$ ✓

(C) Test hypotheses

(D) Measure correlation

Answer: (B) Relate $P(A|B)$ to $P(B|A)$

Authenticity: Frequently Repeated

Q7. A p-value below the significance level suggests you should:

(A) Accept the null

(B) Reject the null hypothesis ✓

(C) Ignore the result

(D) Increase alpha

Answer: (B) Reject the null hypothesis

Authenticity: Frequently Repeated

Q8. The Central Limit Theorem says the sampling distribution of the mean:

(A) Is always uniform

(B) Approaches normal as sample size grows ✓

(C) Has zero variance

(D) Equals the population distribution

Answer: (B) Approaches normal as sample size grows

Authenticity: Frequently Repeated

Q9. Pearson's correlation coefficient ranges between:

(A) 0 and 1

(B) -1 and 1 ✓

(C) -inf and inf

(D) 0 and 100

Answer: (B) -1 and 1

Authenticity: Frequently Repeated

Section 4 - Machine Learning

Q1. Overfitting occurs when a model:

(A) Performs poorly on both train and test

(B) Performs well on train but poorly on unseen data ✓

(C) Has too few parameters

(D) Cannot be trained

Answer: (B) Performs well on train but poorly on unseen data

Authenticity: Frequently Repeated

Q2. Which is an unsupervised algorithm?

(A) Logistic Regression

(B) K-means clustering ✓

(C) Decision Tree classifier

(D) Linear Regression

Answer: (B) K-means clustering

Authenticity: Frequently Repeated

Q3. PCA is primarily used for:

(A) Classification

(B) Dimensionality reduction ✓

(C) Clustering labels

(D) Labeling

Answer: (B) Dimensionality reduction

Authenticity: Community Reported

Q4. Best metric for an imbalanced classification problem:

(A) Accuracy

(B) F1-score ✓

(C) Total count

(D) Learning rate

Answer: (B) F1-score

Authenticity: Frequently Repeated

Q5. A confusion matrix shows:

(A) Feature correlations

(B) Counts of TP/TN/FP/FN ✓

(C) Model weights

(D) Learning curves

Answer: (B) Counts of TP/TN/FP/FN

Authenticity: Community Reported

Q6. Typical loss for regression:

(A) Binary cross-entropy

(B) Mean Squared Error ✓

(C) Hinge loss

(D) Categorical cross-entropy

Answer: (B) Mean Squared Error

Authenticity: Community Reported

Q7. Common hidden-layer activation in deep networks:

(A) Sigmoid

(B) ReLU ✓

(C) Linear

(D) Step

Answer: (B) ReLU

Authenticity: Community Reported

Q8. Gradient descent is used to:

(A) Increase the loss

(B) Minimize the cost function ✓

(C) Split data

(D) Encode features

Answer: (B) Minimize the cost function

Authenticity: Frequently Repeated

Q9. A common way to handle missing values is:

(A) Encoding

(B) Imputation ✓

(C) Normalization

(D) Tokenization

Answer: (B) Imputation

Authenticity: Community Reported

Section 5 - CS Fundamentals

Q1. Which data structure follows FIFO?

(A) Stack

(B) Queue ✓

(C) Tree

(D) Graph

Answer: (B) Queue

Authenticity: Community Reported

Q2. Binary search runs in:

(A) $O(n)$

(B) $O(\log n)$ ✓

(C) $O(n \log n)$

(D) $O(1)$

Answer: (B) $O(\log n)$

Authenticity: Frequently Repeated

Section 6 - SQL

Problem: Second Highest Salary

Find the second highest distinct salary; NULL if it does not exist.

```
SELECT MAX(salary) AS second_highest
FROM employee
WHERE salary < (SELECT MAX(salary) FROM employee);
```

Expected Output

```
second_highest
60000
```

Section 7 - Python

Problem 1: Most Frequent Word

Difficulty: *Easy* | Pattern: *Hash Map* | Authenticity: *Community Reported*

Problem Statement

Return the most frequent word in a sentence; ties resolved by first appearance; case-insensitive.

Input Format

A single line sentence.

Output Format

The most frequent word (lowercase).

Constraints

Words separated by single spaces.

Sample Input

```
the cat sat on the mat the cat
```

Sample Output

```
the
```

Python Solution

```
def most_frequent_word(sentence):
    counts = {}
    best_word, best = None, 0
    for w in sentence.lower().split():
        counts[w] = counts.get(w, 0) + 1
        if counts[w] > best:
            best, best_word = counts[w], w
    return best_word

print(most_frequent_word('the cat sat on the mat the cat'))
```

Step-by-Step Explanation

Count words; keep the best, updating only on a strictly higher count so ties keep the first word. 'the' occurs 3 times.

Dry Run

```
the->1(best);cat->1;the->2;the->3 -> the
```

Time Complexity: **O(N)** | Space Complexity: **O(K)**

Tredence

Full-Length Mock Test 1

Applied Role(s):

Analyst - Data Scientist

Online Assessment Preparation Material

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Every question carries an authenticity label. See the legend on the next page.

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Authenticity Legend & How to Read This Document

Every question and problem in this document is labelled with an **authenticity classification** so you know how closely it maps to what candidates have actually reported. Labels are assigned conservatively based on research; authored practice questions are never presented as real exam questions.

Label	Meaning
Verified Previous-Year	The specific problem/pattern is documented from actual reported assessment experiences.
Frequently Repeated	The topic or pattern type is documented as recurring across multiple hiring cycles.
Community Reported	A widely shared, role-relevant pattern reported across placement communities and forums.
Research-Backed Company Style	Authored to match the researched difficulty, style and pattern - practice, not a real past question.

The correct option in every MCQ is highlighted in green and also restated below the question. Study the pattern, not just the answer.